

Ecam Brussels Engineering School

RF Project Report

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Table of Contents

1 Introduction **2**

2 Phase 1 - 2.4GHz broadcasting and IQ acquisition with SDR **3**
2.1 Objective and experimental approach 3
2.2 Preliminary laboratory experiments 3
2.3 2.4 GHz transmission and I/Q acquisition 3
2.4 Phase 1 outcome and scope 4

3 Phase 2 - Beamforming and Source tracking **4**

4 Phase 3 - Antenna Design **4**

5 Conclusion **4**

1 Introduction

Radio direction finding is the process of estimating the direction from which a radio signal arrive. It is an important capability in wireless systems, where locating a transmitter can support monitoring, signal tracking, and a better understanding of the radio environment. This project investigates a practical approach to direction-of-arrival estimation using software-defined radio (SDR) and beamforming.

The system is developed in three phases. First, a signal in the 2.4 GHz band is transmitted and its in-phase and quadrature (I/Q) samples are acquired using an ADALM-Pluto SDR. These samples provide the complex signal data needed for digital processing. Second, beamforming techniques are explored to combine signals received at multiple antenna elements and estimate the direction of the source, with the aim of following changes in its position. Most of the calculation are done with the help of our telecommunications knowledge, Acquierd with Mr Tihion. Finally, an antenna is designed for the system, considering the requirements of operation at the target frequency.

Together, these phases form an end-to-end study, from signal transmission and acquisition to spatial processing and antenna design. The report presents the approach and results of each phase, and discusses the practical constraints encountered when implementing a direction-finding system with SDR hardware.

2 Phase 1 - 2.4GHz broadcasting and IQ acquisition with SDR

2.1 Objective and experimental approach

The first phase establishes the transmission and reception chain required for the later direction-of-arrival experiments. Its purpose is to generate repeatable radio signals, acquire their complex in-phase and quadrature (I/Q) samples with an SDR, and inspect or recover information from those samples. The laboratory exercises provide preliminary experience with radio transmission, receiver configuration, and frequency-domain analysis.

2.2 Preliminary laboratory experiments

In the first laboratory exercise, an ESP32-C3 is configured to broadcast packets using ESP-NOW on Wi-Fi channel 1, whose centre frequency is 2412 MHz. The transmitter selects a PHY rate of 6 Mbit/s and broadcasts a packet every 250 ms. Each packet contains a constant identifier, a sequence number, a sample index, and one value from a small test data set. This provides a repeatable 2.4 GHz radio transmission whose packets can be used to explore reception and signal observation.

The Pluto SDR scripts also use a continuous-wave (CW) signal as a simple frequency reference. In the laboratory setup, the transmitter and receiver local oscillators are set to 433 MHz, while the transmitter's digital tone is offset by 100 kHz. The resulting signal is therefore expected near 433.1 MHz. With a sampling rate and receiver bandwidth of 2 MHz, the captured I/Q buffer can be transformed using a fast Fourier transform (FFT). The frequency of the largest spectral bin gives an estimate of the received tone's offset from the receiver local oscillator.

The second laboratory script extends this measurement with a live spectrum display. Before calculating the FFT, it applies a Hann window to the samples to reduce spectral leakage caused by observing a finite data buffer. The frequency axis is shifted so that zero frequency is at the centre, then translated by the receiver's local-oscillator frequency to display absolute frequency. A marker indicates the largest spectral peak. These exercises introduce the receiver settings and analysis techniques used in the project ; the 433 MHz CW test is a preliminary measurement and is distinct from the project's 2.4 GHz transmission.

2.3 2.4 GHz transmission and I/Q acquisition

The ESP-NOW transmitter is paired with a Pluto SDR receiver implemented as a GNU Radio flowgraph. It is tuned to 2412 MHz and samples at 20 MSample/s. The flowgraph passes the received samples to an IEEE 802.11 physical-layer decoder, then examines the decoded frame for the test packet identifier and expected data values. Sequence numbers also make it possible to report gaps in the received packet stream. This experiment demonstrates packet reception over the air, in addition to the raw spectrum measurements of the CW exercises.

The project also includes a Pluto SDR CW transmitter configured with a 2.4 GHz local oscillator and a 100 kHz digital tone, placing the emitted tone at approximately 2.4001 GHz. A separate basic receiver script tunes to 2.4 GHz, samples at 2 MSample/s,

and prints a captured I/Q buffer. This is a raw acquisition check ; it does not attempt to decode a message.

BPSK waveform and recovery. For the project signal, the BPSK script configures the Pluto SDR transmitter and receiver at a carrier frequency of 2.4 GHz, with a sampling rate of 1 MSample/s. It generates a reproducible frame of 1024 pseudo-random bits. Each bit is mapped to one of two antipodal symbols, -1 or $+1$, and repeated for eight samples. The corresponding symbol rate is

$$R_s = \frac{f_s}{N_s} = \frac{1 \text{ MSample/s}}{8} = 125 \text{ ksymbol/s},$$

where f_s is the sampling rate and N_s is the number of samples per symbol. The resulting complex-valued waveform is transmitted cyclically, allowing the receiver to capture repeated copies of the same known frame.

The receiver acquires complex samples $x[n] = I[n] + jQ[n]$. One initial acquisition is discarded after configuration to avoid using a transition buffer. The received samples are then processed to compensate for a constant phase rotation, integrate over each symbol with an eight-sample rectangular filter, and select the best of the eight possible sample timings. Because the transmitted frame is known, correlation is used to find its alignment in the received buffer and resolve the possible BPSK sign ambiguity. A hard decision on the real part of each recovered symbol produces the estimated bit sequence. The script compares it with the original bits and reports the bit-error rate (BER) ; it also displays the recovered constellation and received spectrum.

2.4 Phase 1 outcome and scope

Together, the laboratory work and project scripts demonstrate how to generate a known RF waveform, configure an SDR receiver, inspect the received spectrum, and recover a digital sequence from I/Q samples. They also provide a repeatable signal-processing chain that can be used to check transmission and acquisition settings. The scripts report measurements such as the detected frequency and BER when run, but no fixed numerical results are included here because they depend on the hardware setup and the individual acquisition.

These experiments use a single receive channel for the BPSK recovery and spectrum measurements. They therefore do not, on their own, estimate the direction of arrival. Spatial phase comparison across multiple receive channels, together with calibration, is required for that task and is addressed in the following phase.

3 Phase 2 - Beamforming and Source tracking

4 Phase 3 - Antenna Design

5 Conclusion